Problem 2: Mars Rover Sample Collection (PDDL)

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Task 3: Plan Generation

Satisficing: Greedy best-first search with FF heuristic. Optimal: A* with LM-cut heuristic (admissible). Planner: Fast Downward.

Scenario	Configuration	Plan length (total actions)	States expanded	Planning time
Original	Satisficing	30	51	0.0009 s
Original	Optimal	25	5,792	0.27 s
Extended	Satisficing	51	79	0.0018 s
Extended	Optimal	40	2,230,060	122.4 s

Task 4: Goal Verification

All plans were simulated step by step. Every action was applicable, and the goal holds in the final state. In the final state of the optimal plan the rover is at Base, both bays are empty, and Rock1 to Rock5 are at Base.

Task 6: Plan Analysis

1. Plan length comparison

The satisficing plans are longer (30 vs 25, and 51 vs 40). Greedy best-first search only follows the heuristic and does not consider path cost, so it accepts the first plan it finds, often carrying one rock per trip. A* with an admissible heuristic always finds the shortest plan, but it expands many more states and takes longer.

2. Storage bays and drill (optimal plan)

Trip 1: Base → SiteB (Rock2 into Bay1) → SiteD (Rock4 into Bay2) → SiteB → Base, deposit both. Trip 2: Base → SiteA → SiteC → SiteE (pick up drill) → SiteF (Rock5 into Bay1) → SiteE → SiteC (Rock3 into Bay2) → SiteA → Base, deposit both. Trip 3: Base → SiteA (Rock1 into Bay1) → Base, deposit. Both bays are used on every trip except the last. The drill is picked up at SiteE on the way to the two hard rocks, just before it is needed.

3. Extended scenario

With Base ↔ SiteB closed, SiteB can only be reached through SiteD. The rover goes Base → SiteA → SiteC → SiteD using the one-way slope, collects Rock4, then goes to SiteB for Rock2. Since it cannot go back from SiteD to SiteC, it returns through SiteD → SiteE (picking up the drill) → SiteC → SiteA → Base. So yes, the one-way route is used.